Experiment No. 06
Logistic Regression, Thresholds and Classification Metrics

Aim:
To implement a **Logistic Regression classifier**, obtain predicted probabilities, study the effect of different classification thresholds on precision and recall, analyze confusion matrices, and use ROC and Precision-Recall curves to select an appropriate classification threshold.

Objective:
After completing this experiment, students will be able to:

1. To understand the concept of **Logistic Regression** for binary classification.
2. To understand the **sigmoid function** and predicted probabilities.
3. To train a Logistic Regression classifier using a dataset.
4. To obtain class probabilities using predict_proba().
5. To understand the concept of a **classification threshold**.
6. To study how changing the threshold affects precision and recall.
7. To construct and interpret a **confusion matrix**.
8. To calculate classification metrics such as precision and recall.
9. To plot and interpret **ROC and Precision-Recall curves**.
10. To select a suitable classification threshold based on the real-world cost of **false positives and false negatives**.

Pseudo Algorithm :

Steps:

1. **Start.**
2. Load the classification dataset.
3. Separate features X and target y.
4. Split the dataset into training and testing sets.
5. Train a Logistic Regression classifier.
6. Obtain predicted probabilities using predict_proba().
7. Store the positive-class probability.
8. Select different thresholds such as:
    - 0.1
    - 0.3
    - 0.5
    - 0.7
    - 0.9
9. Convert probabilities into class predictions using each threshold.
10. Calculate the confusion matrix for each threshold.
11. Calculate precision and recall.
12. Compare the metrics across thresholds.
13. Plot the ROC curve.
14. Plot the Precision-Recall curve.
15. Check the class distribution.
16. Identify whether the dataset is imbalanced.

17. Select a real-world application where false positives and false negatives have different costs.
18. Select an appropriate threshold based on the application.
19. Justify the selected threshold.
20. **Stop.**

Sample Outcome:

- Threshold = 0.1: Very high recall but many false positives.
- Threshold = 0.3: Higher recall with a more acceptable number of false positives.
- Threshold = 0.5: Reasonable default balance.
- Threshold = 0.7: Higher precision but more false negatives.
- Threshold = 0.9: Very conservative positive predictions and low recall.
- For fraud detection, a lower threshold may be preferred because missing fraudulent transactions is costly.
- The final threshold should be selected using the cost of FP vs FN, rather than automatically choosing 0.5.

Sample Observation:

- The Logistic Regression classifier was successfully trained on the given binary classification dataset.
- Predicted probabilities for the positive class were obtained using predict_proba(). Different classification thresholds ranging from 0.1 to 0.9 were then applied.
- It was observed that changing the threshold significantly affected precision, recall, and the confusion matrix. Lower thresholds generally produced higher recall, while higher thresholds generally produced higher precision.
- ROC and Precision-Recall curves were successfully generated to evaluate classifier performance across different thresholds.

Tools / Software Required:

Hardware: Computer/Laptop, Minimum 4 GB RAM

Software: Python 3.x, Jupyter Notebook or Google Colab, Anaconda (optional)

Python Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, Joblib / Pickle

Viva Questions:

1. What is Logistic Regression?
2. Why is Logistic Regression called regression if it is used for classification?
3. What is the sigmoid function?
4. What is a classification threshold?
5. What generally happens to precision and recall when the threshold increases?
6. What is a confusion matrix?
7. What is the difference between precision and recall?
8. What is the ROC curve?
9. Why is the Precision-Recall curve useful for imbalanced datasets?
10. Why shouldn't we always use a threshold of 0.5?